

NeuroShift: Latest 10 IEEE References (2024-2026)

All 10 are IEEE papers | Base: Chamberland IEEE EMBC 2024 | Updated July 2026

All core-10 are IEEE (10.1109) and 2024 or newer. P1 is BASE. Each module maps to an IEEE paper. Novelty: gaze-select + EMG-confirm + Act/Abstain + FAR for assistive homes.

1. Latest Core-10 (all IEEE)

#	Paper	Year	IEEE Venue / Module
P1	Chamberland et al. Visual Scene Understanding for Enhanced EMG Gesture Recognition	2024	IEEE EMBC BASE EMG+scene
P2	Lin/Yang et al. Hybrid Gaze + FEMG assistive robotic system	2024	IEEE TNSRE Gaze select + confirm
P3	Sakthimohan et al. Smart-home assistive system using eye gestures	2024	IEEE ICACCS Eye -> appliances
P4	Dere et al. Lightweight VLM-guided EMG gesture/motor intent	2025	IEEE Sensors J. Vision-guided EMG
P5	Wang et al. Robust myoelectric recognition for reliable HRI	2025	IEEE RA-L EMG reliability/reject
P6	Jeong & Woo. CIDER on-device smart-home intent reasoning	2025	IEEE Access Smart-home intent
P7	TFVF-CNN sEMG + visual feature fusion	2024	IEEE ROBIO Multimodal fusion
P8	Multimodal sEMG + vision hand gesture fusion	2024	IEEE M2VIP Multimodal fusion (alt)
P9	Zhao/Lv et al. RT-DETR: DETRs Beat YOLOs	2024	IEEE/CVF CVPR Object detection
P10	Li et al. Multimodal dynamic fusion intent, assisted older adults	2025	IEEE ICIVP Assistive intent

2. Module map for sir

Our module	IEEE papers	We improve by
Overall idea	P1 BASE	Gestures -> home intention
Object detection	P9 (+ YOLO in P1)	Detect home appliances
Gaze / eye select	P2, P3	Select appliance by looking
EMG confirm	P4, P5	Sparse MyoWare confirm, not gesture dictionary
Fusion	P7, P8, P10	Home Act/Abstain decision
Act/Abstain	P5 (reject/reliability)	FAR metric for false home actuations
Smart-home IoT	P3, P6	Indian wired relay hub + assistive safety

3. IEEE DOIs to download

- P1 BASE: 10.1109/EMBC53108.2024.10782354
- P2: 10.1109/tnsre.2024.3443073
- P3: 10.1109/icaccs60874.2024.10717287
- P4: 10.1109/jsen.2025.3565766
- P5: 10.1109/lra.2025.3546095
- P6: 10.1109/access.2025.3634621
- P7: 10.1109/robio64047.2024.10907446
- P8: 10.1109/m2vip62491.2024.10746196
- P9: RT-DETR, IEEE/CVF CVPR 2024 (arXiv 2304.08069)
- P10: 10.1109/icivp66296.2025.00029

4. Year count

2024: 5 IEEE papers (including base) | 2025: 5 IEEE papers | All IEEE, all 2024+

5. Honest note for guide

- Earlier Act/Abstain paper (Gaus et al.) was arXiv preprint -> REMOVED for IEEE-only list.
- Act/Abstain is now OUR contribution; IEEE reliability/reject basis is P5 (RA-L 2025).
- RT-DETR (P9) is IEEE/CVF CVPR (IEEE Xplore); can be swapped if strictly IEEE-only journal/conf.

6. Novelty line for guide

Latest IEEE papers already study EMG+vision, gaze+FEMG, eye-controlled homes and reliable EMG. NeuroShift's gap: gaze-selected home appliances + sparse EMG confirmation + Act/Abstain, measured by False Actuation Rate for assistive wired-home control.